NVIDIA

Networks for AI

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FORWARD-LOOKING STATEMENTS

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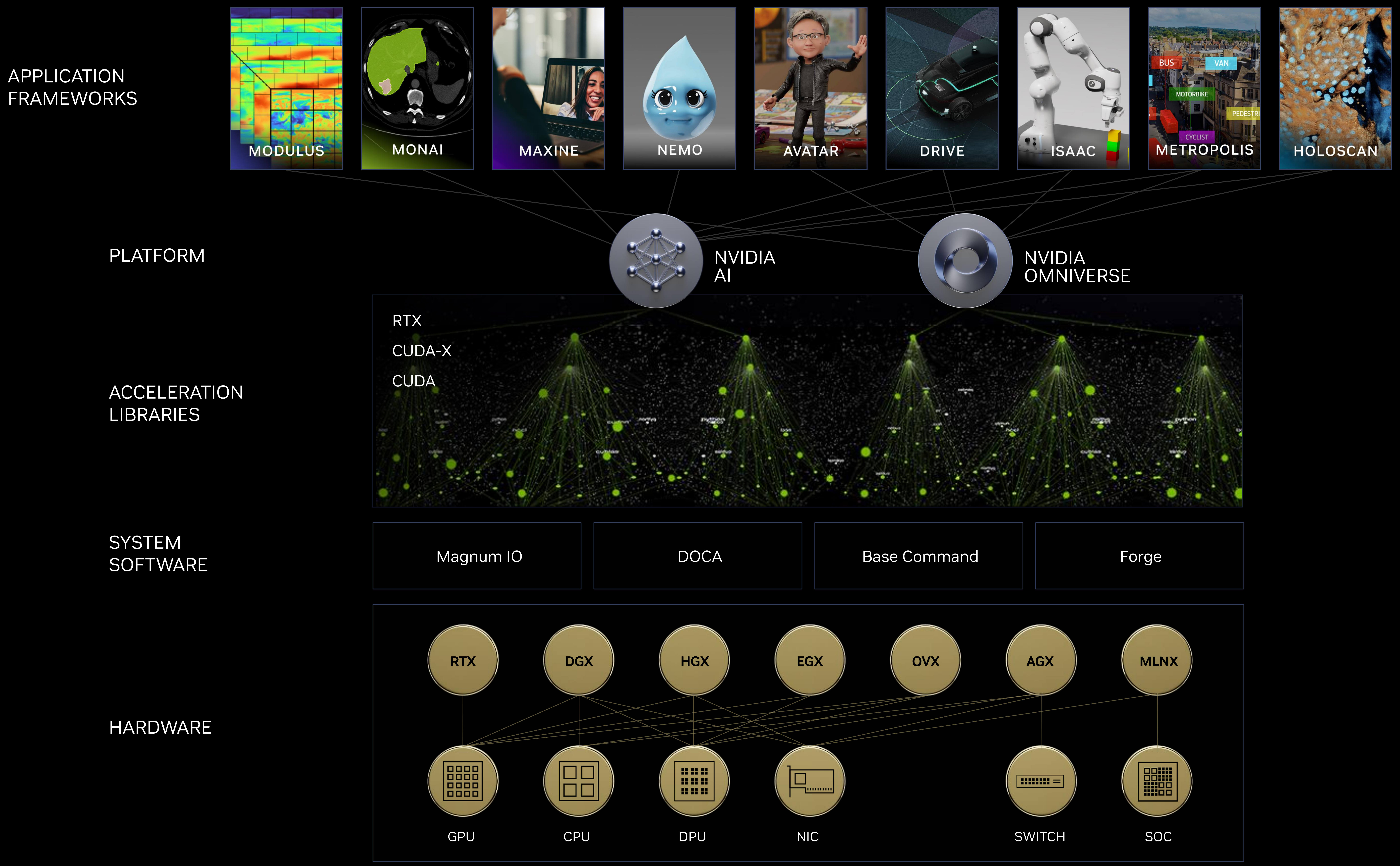
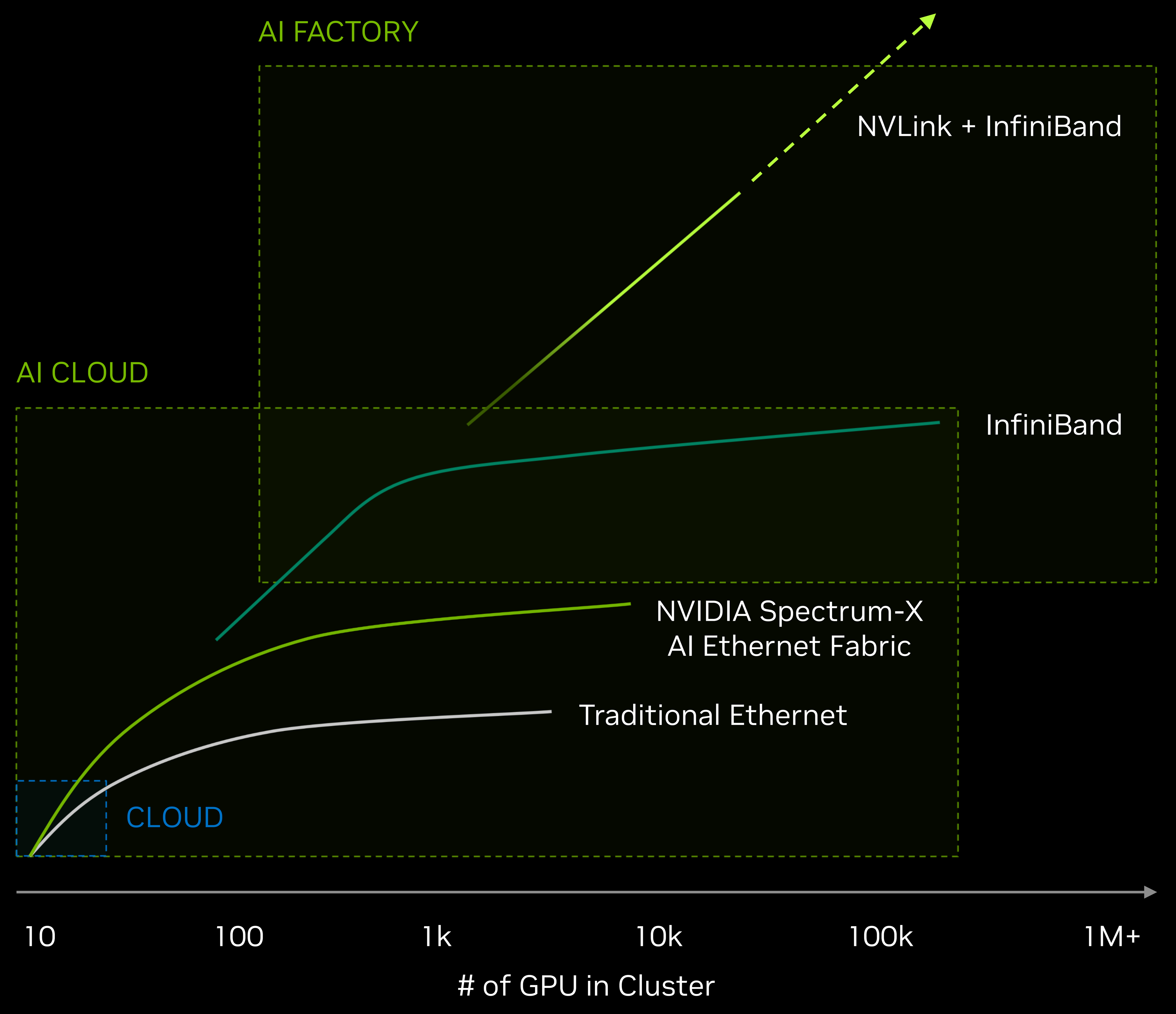

APPLICATION FRAMEWORKS
MODULUS
MONAI
MAXINE
NEMO
AVATAR
DRIVE
ISAAC
BUS
VAN
MOTORBIKE
PEDESTRI
CYCLIST
METROPOLIS
HOLOSCAN
PLATFORM
NVIDIA AI
NVIDIA OMNIVERSE
ACCELERATION LIBRARIES
RTX
CUDA-X
CUDA
SYSTEM SOFTWARE
Magnum IO
DOCA
Base Command
Forge
HARDWARE
RTX
DGX
HGX
EGX
OVX
AGX
MLNX
GPU
CPU
DPU
NIC
SWITCH
SOC

The Data Center is The Computer

The network defines the data center



- Cloud
  - Multi-tenant
  - Variety of small-scale workloads
  - Traditional ethernet network can suffice
- Generative AI Cloud
  - Multi-tenant
  - Variety of workloads including larger scale Generative AI
  - Traditional ethernet network for North-South traffic
  - NVIDIA Spectrum-X ethernet for AI fabric (East-West)
- AI Factories
  - Single or few users
  - Extremely large AI models
  - NVIDIA NVLink and InfiniBand gold standard for AI fabric

The NVIDIA AI Network Advantage

Hardware & software accelerated in-network computing

Software Acceleration

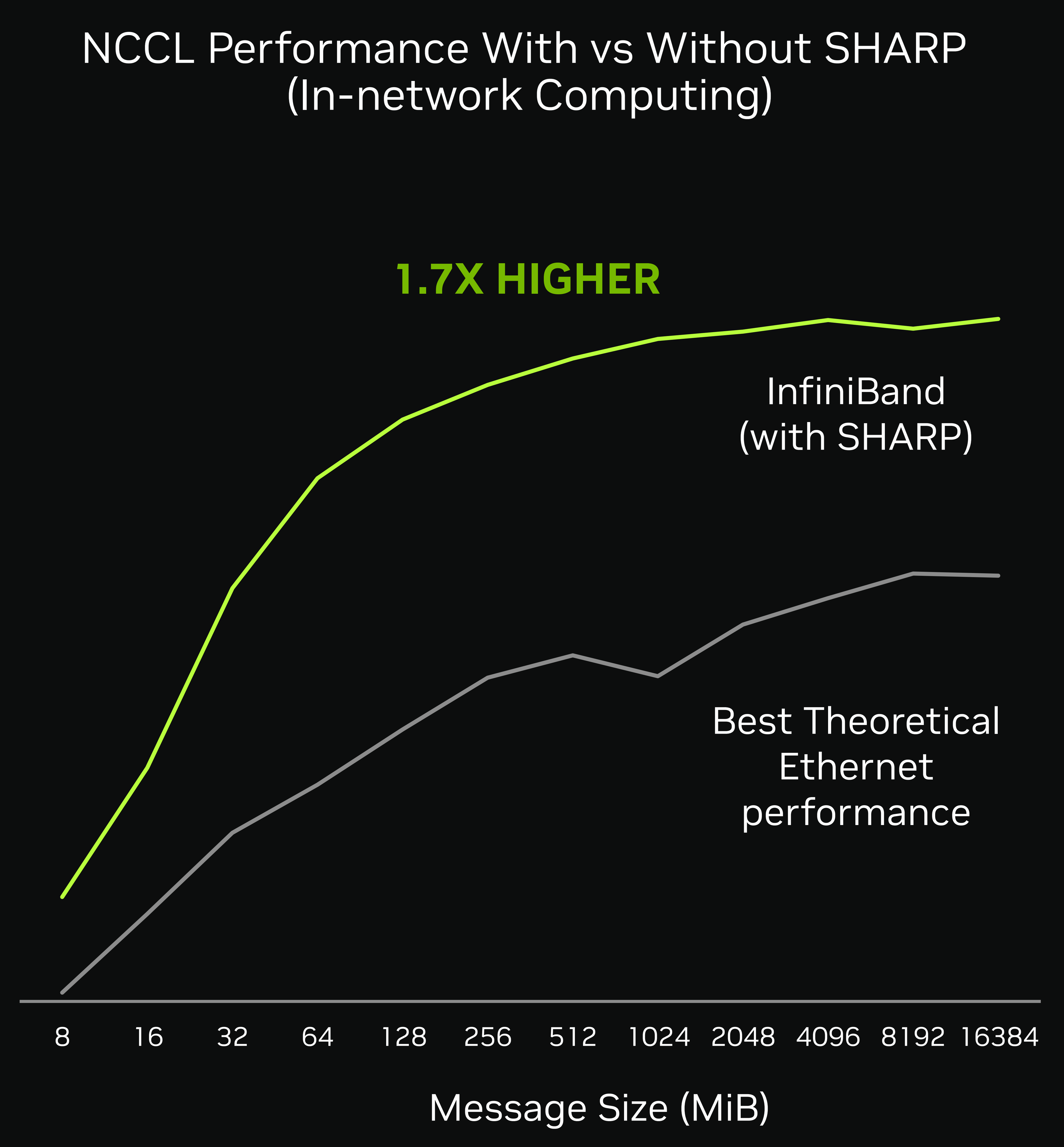
NCCL — NVIDIA Collective Communication Library

The SDK library for AI communications - connects the GPUs and the network for the AI network operations.

Hardware Acceleration

SHARP — NVIDIA Scalable Hierarchical Aggregation and Reduction Protocol Technology

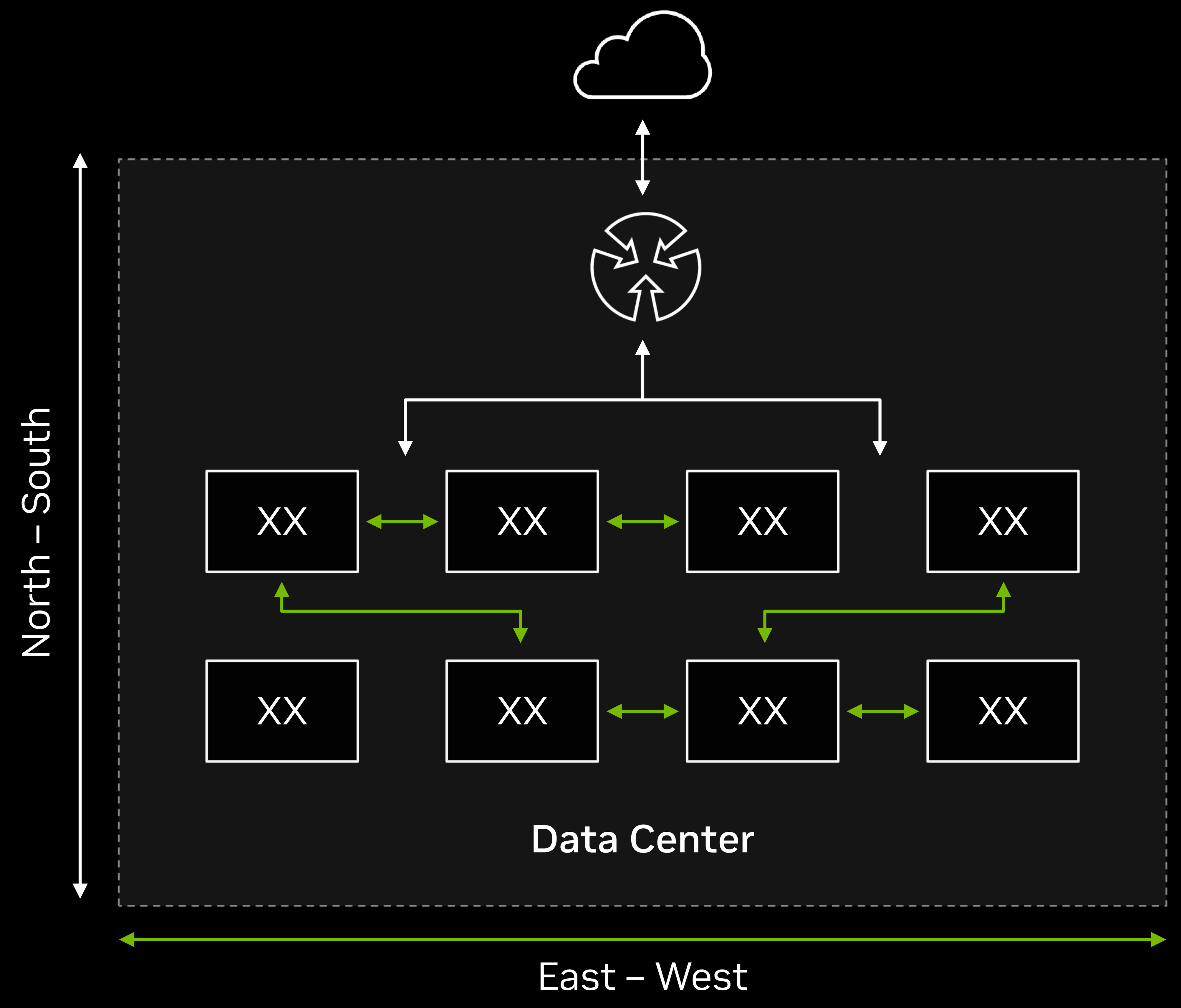
SHARP is part of the InfiniBand and NVLink switch ASICs. It enables the network to perform data reduction operations, an important element of AI workloads. This decreases the amount of data traversing the network and dramatically reduces collective operations time.



In the Cloud — The Two Worlds of Ethernet

East-West for distributed and disaggregated processing | North-South for user-to-cloud communications

Control / User Access Network (North-South) Traditional Ethernet	AI Fabric (East-West) Spectrum-X
Loosely Coupled Applications	Distributed Tightly-Coupled Processing
TCP (Low Bandwidth Flows and Utilization)	RoCE (High Bandwidth Flows and Utilization)
High Jitter Tolerance	Low Jitter Tolerance (Long Tail Kills Performance)
Heterogeneous Traffic Average Multi-Pathing	Bursty Network Capacity Predictable Performance

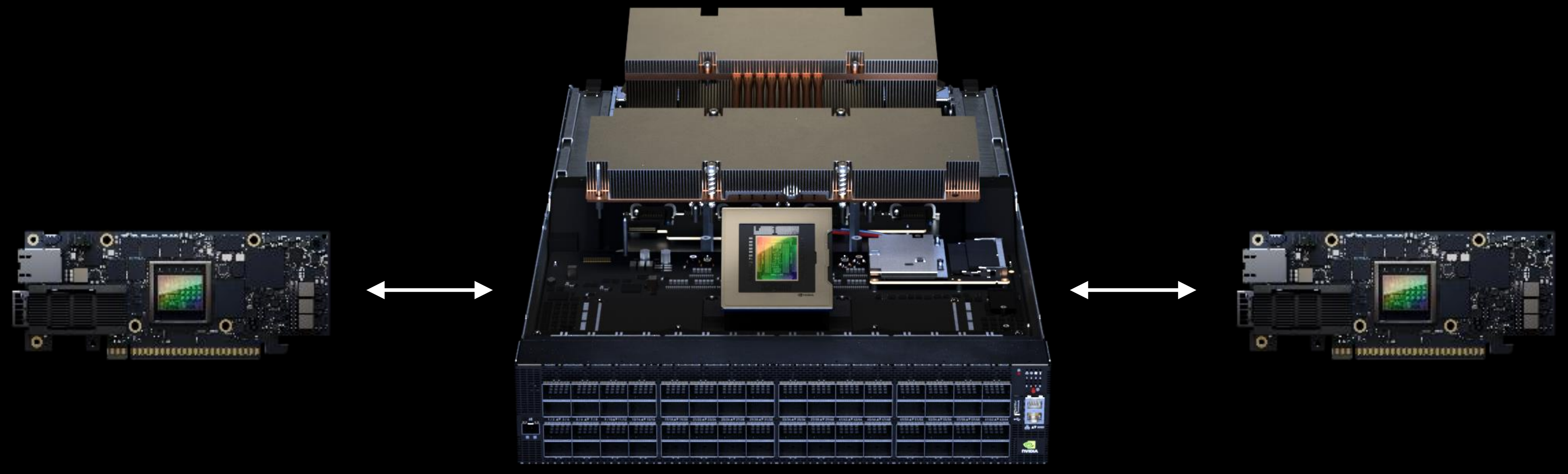


NVIDIA Spectrum-X Ethernet AI Platform

The ethernet fabric for AI

Spectrum-4 Switch

100 billion transistors, TSMC 4N
51.2T bandwidth, 100G SerDes
64 X 800G Ports, 128 X 400G ports
8K GPUs in 2-level, 500K in 3 level



BlueField-3 DPU | Spectrum-4 Switch | BlueField-3 DPU

SONiC	Cumulus	NetQ	DOCA Services	Air
SAI/SPSDK	DOCA	Magnum IO / NCCL		

Architectual Advantages over Traditional Ethernet…

- End-to-end optimized (DPU to Switch)
- Adaptive routing over lossless ethernet
- Congestion control for multi-tenant traffic isolation

…Deliver 1.6X Better AI Fabric Performance vs Traditional Ethernet

- 95% effective bandwidth vs 60% for traditional ethernet
- Maintains NCCL performance in noisy environment
- Secured virtualized network

New class of ethernet, built from the ground-up for Generative AI cloud workloads

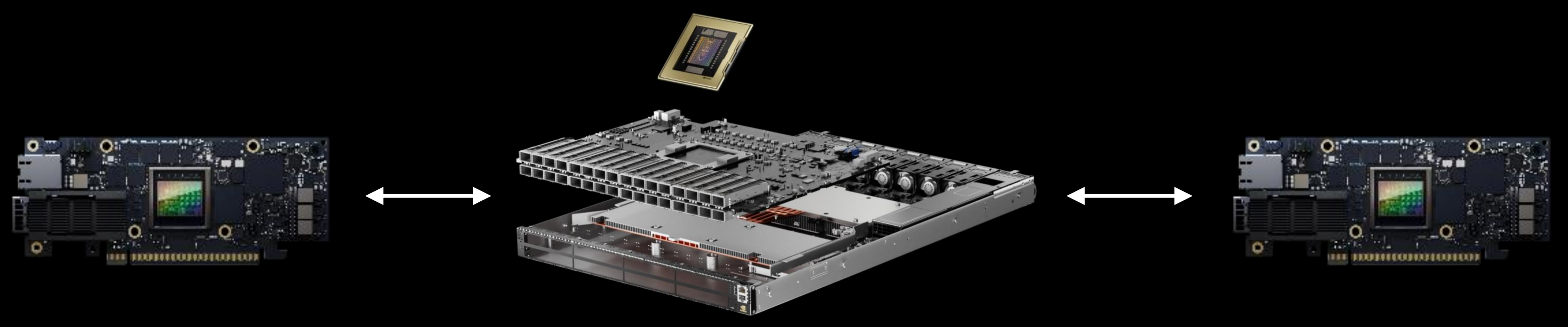
NVIDIA

NVIDIA Quantum InfiniBand Platform

The gold standard for large scale AI

Quantum-2 Switch

57 billion transistors, TSMC 7N
64 X 400G ports, 100G SerDes
64 In-Network Computing AI Engines
65K GPUs in 3-level, 2M in 4-level



BlueField-3 DPU

Quantum-2 Switch

BlueField-3 DPU

NVOS	UFM	DOCA Services	Air
DOCA		Magnum IO / NCCL	

InfiniBand's Inherent Architectural Differentiators…

- Architected for large scale, end-to-end optimized
- In-network computing with SHARP
- Pure software-defined network

…Deliver >2X AI Fabric Performance over Traditional Ethernet

- Limitless GPU scaling (3-level 65K GPUs, 4-level 2M GPUs)
- 1.7X higher NCCL AllReduce bandwidth performance
- Nearly 100% effective bandwidth at scale
- Extremely low latency, 10x better at scale and under load

Hardened over 20+ years for the most extreme-scale workloads in the world's largest supercomputers

The Network “Pays for Itself”!

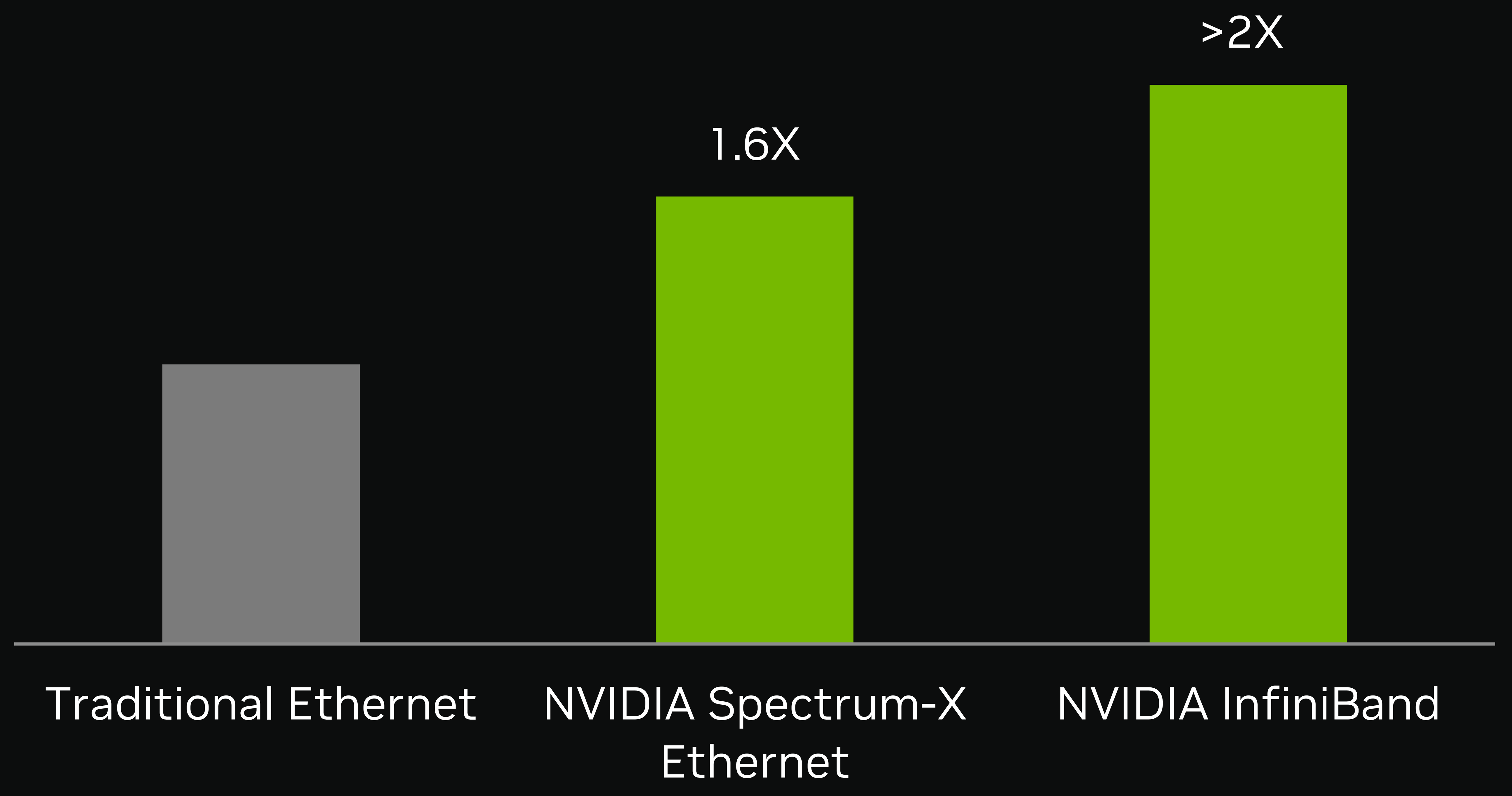
Performance gains far outweigh the entire cost of the network

NDR 400G InfiniBand (64-port switches)

3 switch level network topology (up to 65K nodes)
4 switch level network topology (up to 2M nodes)

InfiniBand offers the most scalability...

Relative AI Fabric Performance
(NCCL Allreduce)

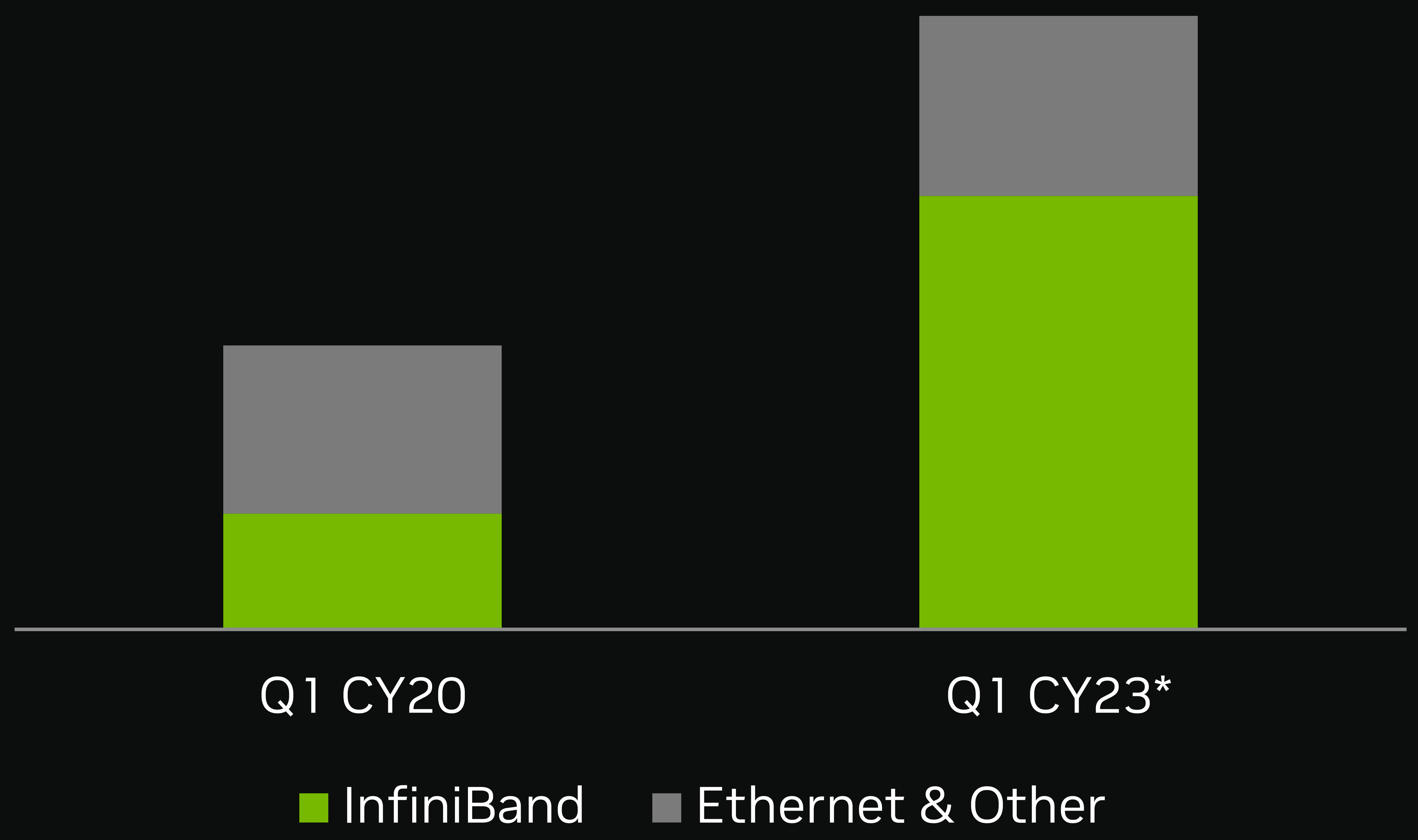


... and the fastest performance

Strong Growth and Large Opportunity

Data Center Networking Growth Turbocharged by AI

Revenue more than doubled since Mellanox acquisition
NVIDIA InfiniBand revenue more than tripled
Spectrum-X to boost cloud AI ethernet market



NVIDIA Networking Revenue

**Q1 CY23 refers to Q1 of NVIDIA's FY24, ending April 30, 2023. The Mellanox acquisition closed on April 27, 2020.*

Data Processing Units — In every server
Data Center Switching — InfiniBand & ethernet
High speed optical modules & cables

$60 BILLION

Networking Market Opportunity

NVIDIA